include the libraries needed for this project.

- ***NTPClient.h*** is time library which does graceful NTP server synchronisation.
- ***ESP8266WiFi.h*** library provides ESP8266-specific Wi-Fi methods we are calling to connect to network.
- ***WiFiUdp.h*** library handles UDP protocol, like opening a UDP port, sending and receiving UDP packets, etc.

Next, we set up a few constants

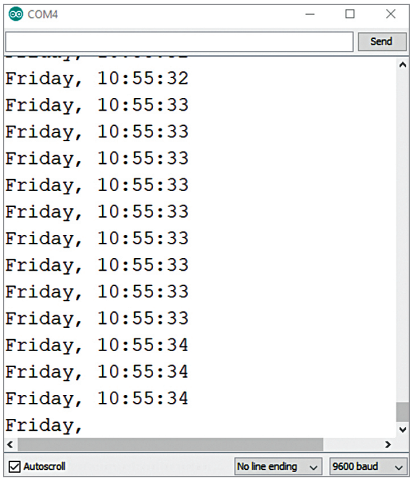

Fig. 6: Serial monitor fetching time from NTP

like SSID, Wi-Fi password, and UTC offset that you are already aware of. We also define *daysOfTheWeek2D* array.

Before initialising NTP Client object, we need to specify the address of the NTP server we wish to use:

```
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP, "pool.ntp.org", utcOffsetInSeconds);
```

The command *pool.ntp.org* automatically picks time servers that are geographically close to you. But if you want to choose explicitly, use one of the sub-zones of the command mentioned below.

Area	HostName
Worldwide	pool.ntp.org
Asia	asia.pool.ntp.org
Europe	europe.pool.ntp.org
North America	north-america.pool.ntp.org
Oceania	oceania.pool.ntp.org
South America	south-america.pool.ntp.org

Once ESP8266 is connected to the network, initialise the NTP client using begin()function, like:

```
timeClient.begin();
```

Now we can simply call the update()function whenever we want current day and time. This function

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
D1, D2	- 1N4007 rectifier diode
Resistor (1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
Capacitors:	
C1	- 1000µF, 25V electrolytic
C2	- 100µF, 16V electrolytic
Miscellaneous:	
CON1	- 2-pin connector
X1	- 230V AC primary to 12V-0-12V, 750mA secondary transformer
LCD1	- LCD LM016L
	- NodeMCU V3
	- Flexible wire

transmits a request packet to an NTP server using UDP protocol and parses the received time stamp packet into a readable format, like:

```
timeClient.update();
```

You can access the day and time information by accessing the NTP Client library functions, as shown in Fig. 6.

After installing all above-mentioned libraries properly, upload the source code *NODE_EFY_ARTICLE.ino*. Now connect the components according to circuit diagram. Power the circuit using 5V power supply described earlier, or you can use your phone charger adaptor that gives an output of 5V.

Wait for a few seconds and you can see the time on display. The best part of this project is that you need not set the time as it automatically takes the right time from NTP.

Assemble the circuit based on circuit diagrams shown in Fig. 3 and Fig. 4. Connect the primary of transformer to 230V AC and its secondary to diodes D1 and D2. The final product (prototype) is shown in Fig. 7. EFY

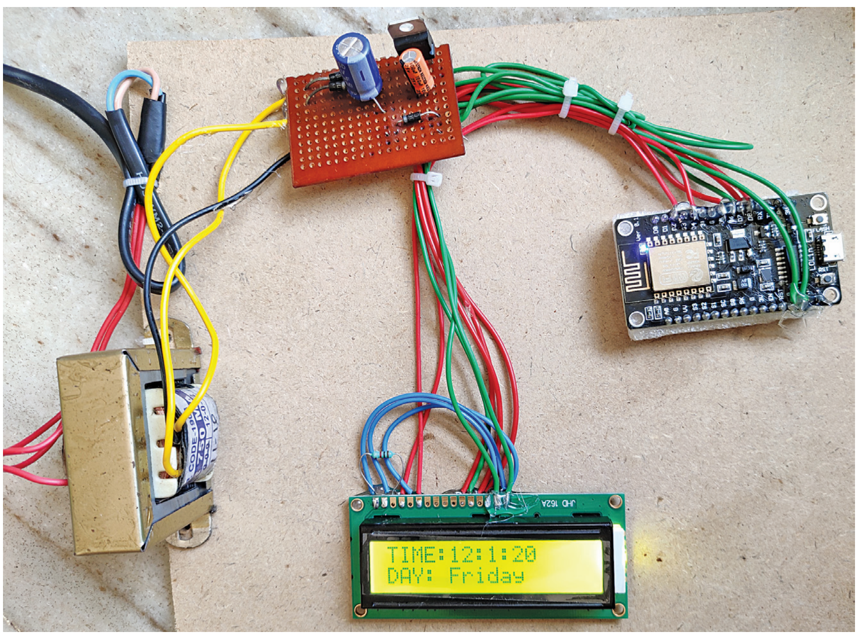

Fig. 7: Author's prototype

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